LAB 1: REQUIREMENTS ENGINEERING

Requirements Table | Problem Statement #37

Group Travel Expense Simplification Engine

Scope: multi-currency group expense recording, daily-rate conversion, and minimum-transaction debt settlement

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall compute the optimal debt settlement graph using min-flow algorithms to minimize total peer-to-peer payment transactions.	High	Pass: Group balance settles to net zero with minimal payment edges. Fail: Sum of debts does not equal sum of credits.	Core purpose of the system and directly specified by the problem statement.	Given requirement retained as specified.
FR-002	Functional	The system shall allow a Group Traveler to record an expense by entering the amount, currency, payer, and participating travelers.	High	Pass: A valid expense is saved with amount, currency, exactly one payer, and at least one participant. Fail: Any required field is missing or invalid.	Accurate expense capture is required before balances and settlements can be computed.	Supports uneven group expenses.
FR-003	Functional	The system shall convert each foreign-currency expense to the trip settlement currency using the applicable daily exchange rate.	High	Pass: For every supported foreign-currency expense, the converted amount equals the stored source amount multiplied by the retrieved daily rate, subject to documented rounding. Fail: No applicable rate is used.	The problem explicitly requires multi-currency conversion using daily exchange rates.	Base-currency expenses may use a 1:1 conversion rate.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-004	Functional	The system shall calculate each traveler's net balance from all recorded expenses and their assigned shares.	High	Pass: For a completed calculation, the sum of all traveler net balances is zero, subject to rounding tolerance. Fail: The total is non-zero beyond the stated tolerance.	Net balances are the input to the minimum-transaction settlement graph.	Enables verification before settlement generation.
FR-005	Functional	The system shall display the generated settlement plan as a list of payer-to-payee transactions with the amount of each payment.	High	Pass: Every transaction identifies one payer, one payee, and a positive amount, and executing all listed transactions reduces every net balance to zero within rounding tolerance. Fail: Any balance remains unsettled.	Travelers need an actionable representation of the optimized debt graph.	Makes the algorithm output usable by the group.
NFR-001	Nonfunctional	The settlement calculation engine must execute in under 100 ms for groups of up to 50 travelers with 500 expenses.	High	Pass: Benchmarking tests confirm settlement calculation latency below 100 ms for the specified workload under simulated peak load. Fail: Any measured calculation exceeds 100 ms.	Performance is explicitly specified in the problem statement and supports responsive use.	Given requirement retained as specified.
NFR-002	Nonfunctional	The system shall restrict trip expense and settlement data to authenticated group members and Trip Admins and protect data in transit using TLS 1.2 or higher.	High	Pass: 100% of unauthorized-access test cases are denied and all application connections carrying trip data use TLS 1.2 or higher. Fail: Any unauthorized request succeeds or any trip-data connection uses unencrypted transport.	Expense and settlement information is shared group data and must be protected from unauthorized access.	Security requirement is measurable through access-control and transport-security tests.

LAB 1: USE-CASE FLOW SPECIFICATION

UC-05: Generate Optimal Settlement Plan

Group Travel Expense Simplification Engine

Primary Actor(s)	Trip Admin; Group Traveler
Supporting Actor	Daily Exchange Rate Provider
Preconditions	A trip exists; participating travelers are registered; expenses are recorded; required daily exchange rates are available; recorded expenses are valid.
Postconditions	All traveler net balances are calculated; an optimized payer-to-payee settlement plan is generated; total net balance is zero within the defined rounding tolerance.

Main Success Scenario

1. The actor selects “Generate Settlement Plan” for the trip.

2. The system retrieves all valid recorded expenses for the trip and identifies the applicable settlement currency.

3. For each expense in a foreign currency, the system obtains the applicable daily exchange rate from the Daily Exchange Rate Provider and converts the amount to the settlement currency.

4. The system calculates each traveler’s total paid amount, total share, and resulting net balance.

5. The system verifies that the sum of all net balances is zero within the defined rounding tolerance.

6. The system applies the min-flow settlement algorithm to match travelers with positive and negative net balances while minimizing the number of payment transactions.

7. The system generates a settlement plan containing each payer, payee, and payment amount.

8. The system displays the settlement plan and its total transaction count to the actor. The use case ends successfully.

Alternate Flow: Daily Exchange Rate Unavailable

3a. The system requests the daily exchange rate for a foreign-currency expense, but the Daily Exchange Rate Provider does not return a valid rate.

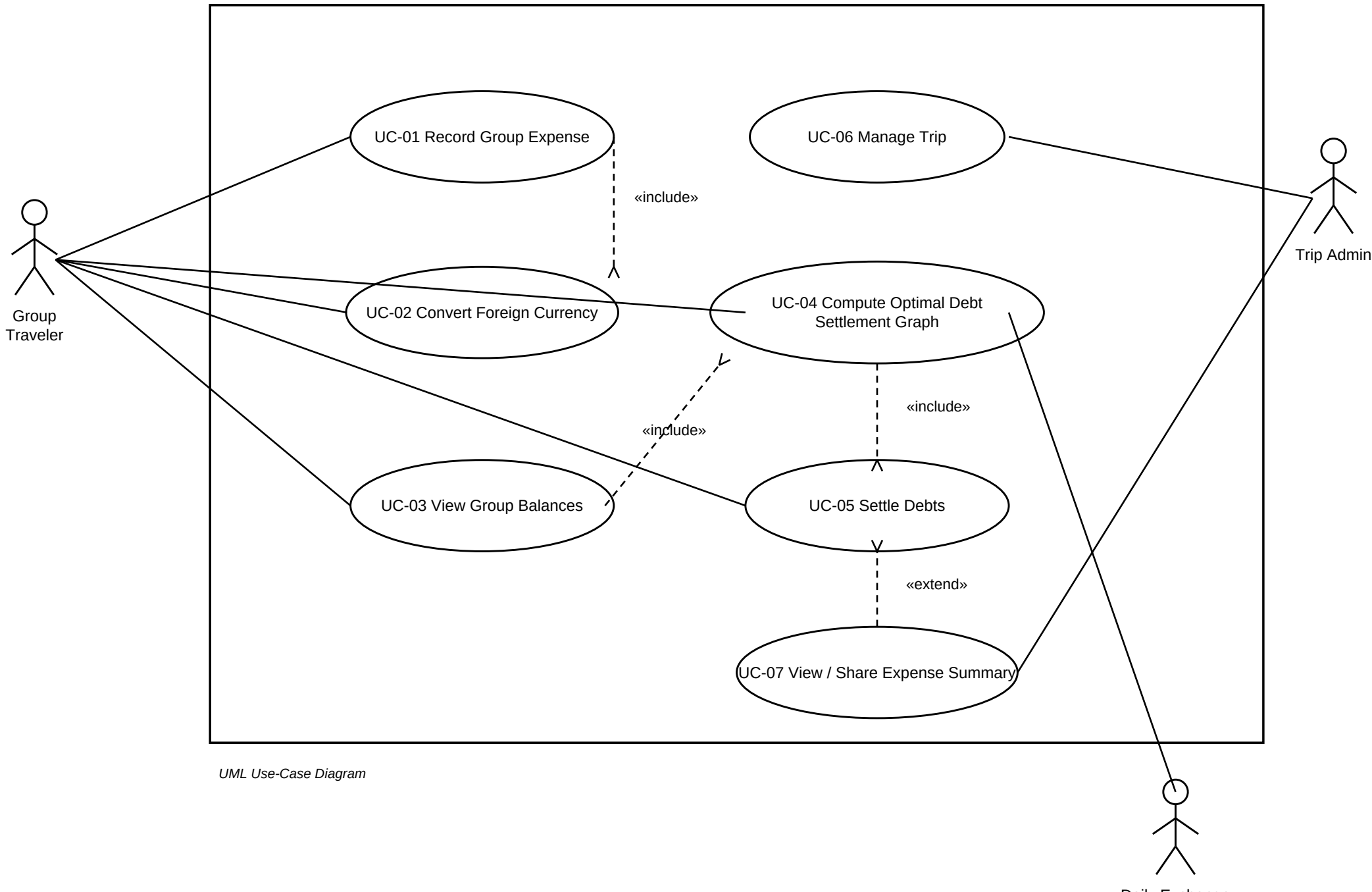
3a1. The system identifies the affected currency and informs the actor that the settlement plan cannot be calculated with the missing rate.

3a2. The system does not generate or display a final settlement plan using an assumed or stale rate.

3a3. The actor may retry the calculation after a valid daily rate becomes available. The use case terminates without creating an invalid settlement plan.

Traceability: UC-05 operationalizes FR-001, FR-003, FR-004, and FR-005 and reflects the multi-currency expense-sharing and minimum-transaction settlement scope.

UML Use-Case Diagram



UML Use-Case Diagram